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Form Approved
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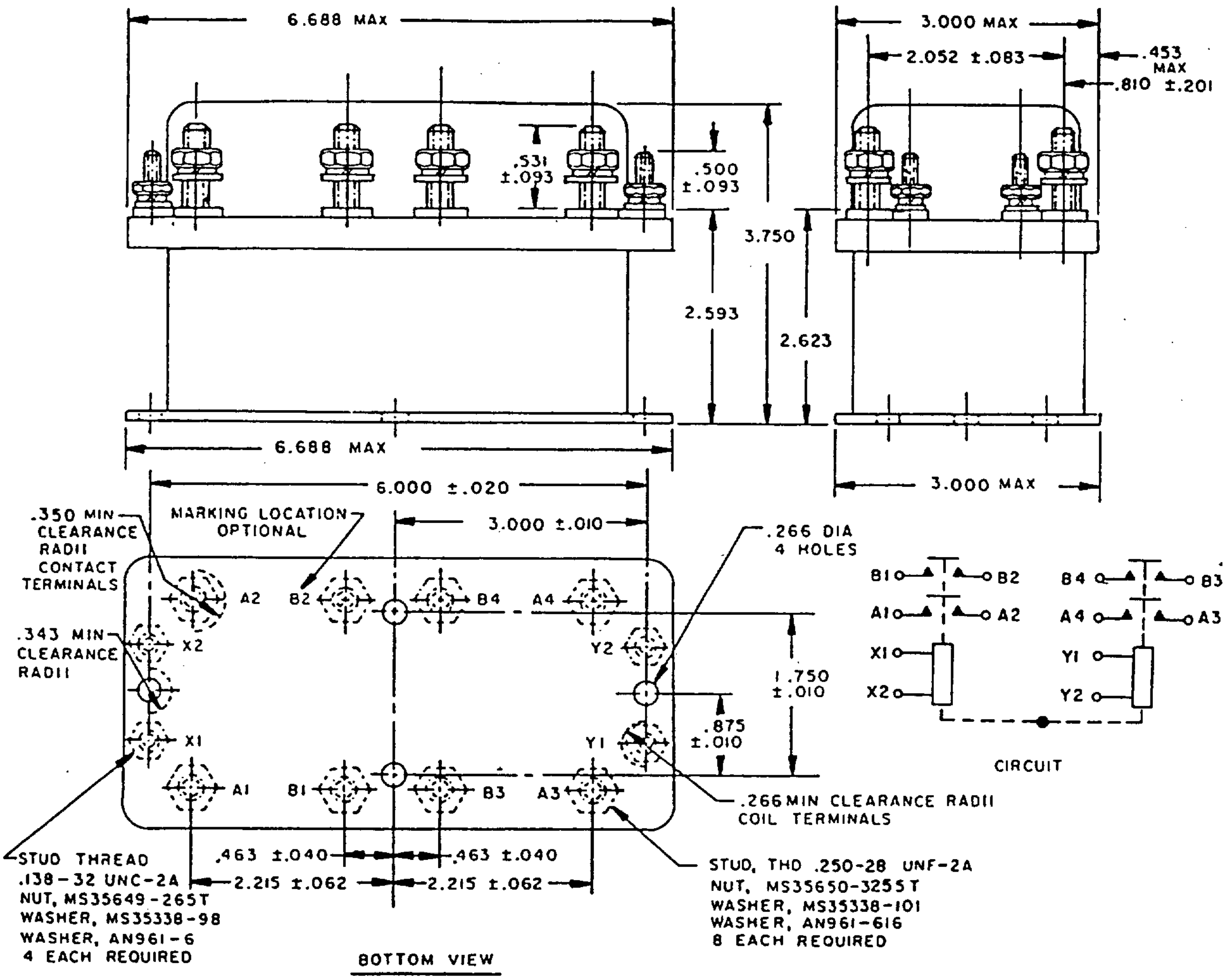
INACTIVE FOR NEW DESIGN AFTER 28 September 1990
NO SUPERSEDING STANDARD


6.688 MAX
3.000 MAX
2.052 ±.083
.453 MAX
.810 ±.201
.531 ±.093
.500 ±.093
3.750
2.593
2.623
6.688 MAX
3.000 MAX
6.000 ±.020
3.000 ±.010
.350 MIN CLEARANCE RADII CONTACT TERMINALS
MARKING LOCATION OPTIONAL
.266 DIA 4 HOLES
.343 MIN CLEARANCE RADII
A2
B2
B4
A4
X2
Y2
X1
Y1
A1
B1
B3
A3
1.750 ±.010
.875 ±.010
.266 MIN CLEARANCE RADII COIL TERMINALS
B1
B2
A1
A2
X1
X2
B4
B3
A4
A3
Y1
Y2
CIRCUIT
STUD THREAD .138-32 UNC-2A
NUT, MS35649-265T
WASHER, MS35338-98
WASHER, AN961-6
4 EACH REQUIRED
.463 ±.040
.463 ±.040
2.215 ±.062
2.215 ±.062
STUD, THD .250-28 UNF-2A
NUT, MS35650-3255T
WASHER, MS35338-101
WASHER, AN961-616
8 EACH REQUIRED
BOTTOM VIEW

THIS SPECIFICATION IS APPROVED FOR USE BY all Departments and Agencies of the Department of Defense.

PREPARING ACTIVITY: Air Force - 85
CUSTODIANS: **ARMY** - **NAVY** - AS
AIR FORCE - **DLA** -

REVIEW:
USER:
PROJECT NUMBER: 5945-0833-01

MILITARY SPECIFICATION SHEET

TITLE RELAY, ELECTROMAGNETIC, 100 AMPERES, 2PDT, N.O. TYPE II, NONHERMETICALLY SEALED, MECHANICALLY INTERLOCKED

SPECIFICATION SHEET NUMBER
MS25031L 28 Sep 1990
SUPERSEDING
MS25031K 31 AUG 82
AMSC N/A **FSC** 5945

DISTRIBUTION STATEMENT
A. Approved for public release; distribution is unlimited.

DD Form 672, MAY 88 *Previous editions are obsolete.*

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Inches	mm	Inches	mm
.010	0.25	.500	12.70
.020	0.51	.531	13.49
.040	1.02	.810	20.57
.062	1.57	.875	22.23
.083	2.11	1.750	44.45
.093	2.36	2.052	52.12
.138	3.51	2.215	63.88
.201	5.11	2.593	65.86
.250	6.35	2.623	66.62
.266	6.76	3.000	76.20
.343	8.71	3.750	95.25
.350	8.89	6.000	152.40
.463	11.51	6.688	169.88

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.032 (0.81 mm).
4. Additional flat washer may be used for terminal seat.
5. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.
7. Terminal numbers shall not appear on relay header. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location.
8. Mechanical linkage shall be provided to prevent both relays being closed at the same time.

TABLE I. Dash numbers and characteristics.

Dash number MS25031-	Type	Coil	Terminal type	Mounting	Maximum weight in pounds
D1B	II	DC	Stud	Plate	3.5

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TABLE II. Operating characteristics.

MS part number MS25031-	Coil data											Time - (milliseconds maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop-out voltage 2/	Operate 3/	Release 4/	Bounce			
		Volts 1/	Freq. Hz	Res Ω	Volts	Amperes	Normal 2/	High temp test	Cont current test					Main		Aux	
														NO	NC	NO	NC
D1B	X1,X2 Y1,Y2	28	DC	47.5	29	.70	18	21	22.5	7.0	1.5	35	15	3.5	---	---	---

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

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TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/			
		Main		Aux		Main		Aux		Main		Aux	
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz
Resistive	50	100				75							
Inductive	10	80											
Inductive													
Motor	50	80				65							
Lamp													
Transfer load 2/													
Mechanical life reduced current	100	25				25							
Intermediate current		Applicable in accordance with MIL-R-6106											

1/ Absence of value indicates relay is not rated for 3 phase application.

2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Environmental characteristics

Temperature range	-55°C to +71°C
Maximum altitude rating	50,000 ft
Shock G-level	25 g's
Duration	6-9 ms
Max duration contact opening	2 ms
Vibration - sinusoidal (see chart below)	
G-level	g
Frequency range	Hz
Vibration - random	
Applicable spec	N/A
Power spectral density	N/A
RMS G min	
Frequency range	N/A
Curve	
High shock	N/A
Acceleration	10 g's

Electrical characteristics

Min insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms
Dielectric strength (sea level)	2-5 seconds

	Initial		After life tests	
	28 V dc	115 V ac	28 V	115 V
Coil to case	1,250	N/A	1,000	N/A
Aux contacts	1,250	N/A	1,000	N/A
All other points	1,250	1,500	1,000	1,125

Dielectric strength (altitude) 1 minute

	28 V dc	115 V ac
Coil to case	500	N/A
Aux contacts	500	N/A
All other points	500	500

Max contact drop initial	.150 volt
After life test	.175 volt
Overload current (N.O.)	800 amperes
Rupture current (N.O.)	1,000 amperes
Duty rating	Continuous
RFI specification (Applicable to coil circuits of ac operated relays)	

Quality conformance inspection

Performance of groups B and C tests are not applicable.

Group A acceptance reports shall be submitted to the preparing activity on a yearly basis in order to retain qualification for this military standard sheet.

Dash number MS25031-	Vibration level				
	5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1500 Hz
D1B	.08 DA	.06 DA	2 g's	2 g's	

PREPARING ACTIVITY: Air Force - 85 **CUSTODIANS:** **ARMY** - **NAVY** - AS **AIR FORCE** - **DLA** - **REVIEW:** **USER:** **PROJECT NUMBER:** 5945-0833-01	**MILITARY SPECIFICATION SHEET** **TITLE** RELAY, ELECTROMAGNETIC, 100 AMPERES, 2PDT, N.O. TYPE II, NONHERMETICALLY SEALED, MECHANICALLY INTERLOCKED	**SPECIFICATION SHEET NUMBER** **MS25031L** **SUPERSEDING** MS25031K 31 AUG 82 **AMSC** N/A **FSC** 5945
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